Daisyworld with an Ocean, part 2: adding life

EC2113 Intro to Climate · Ocean module · app: joymonteiro.github.io/intro_climate_science/tutorials/daisyworld-ocean

This handout goes with the same app as part 1, "Daisyworld with an Ocean", and uses its presets 4 to 7. Part 1 built a bare planet: land, a mixed layer and a deep ocean, pushed by random weather. Here, daisies grow on the land and grazers eat them. Open the app next to this handout.

- **Part 1** explains what life adds to the model, draws the feedbacks as a causal loop diagram, and says what every control does.
- **Part 2** walks through six runs. For each one it says what to set, asks you to predict what will happen, and then explains what you see. Try to answer each "Before you look" question before you read on.
- **Part 3** sums up what to take away, and says what the model leaves out.
- **Take it further** is a spreadsheet exercise for after the tutorial.

You need no maths for this handout.

> **The point in two sentences.** On their own, daisies steady the land's temperature: they form loops that push back against any change. Add grazers and the ocean's slow memory, and the push-back comes too late, so the planet overshoots into a boom–bust cycle; add weather, and the ocean's memory sets the size of each bust, so a planet driven by nothing slow swings over decades and centuries.

Part 1. Life, its feedbacks, and the controls

1.1 What life adds, in words

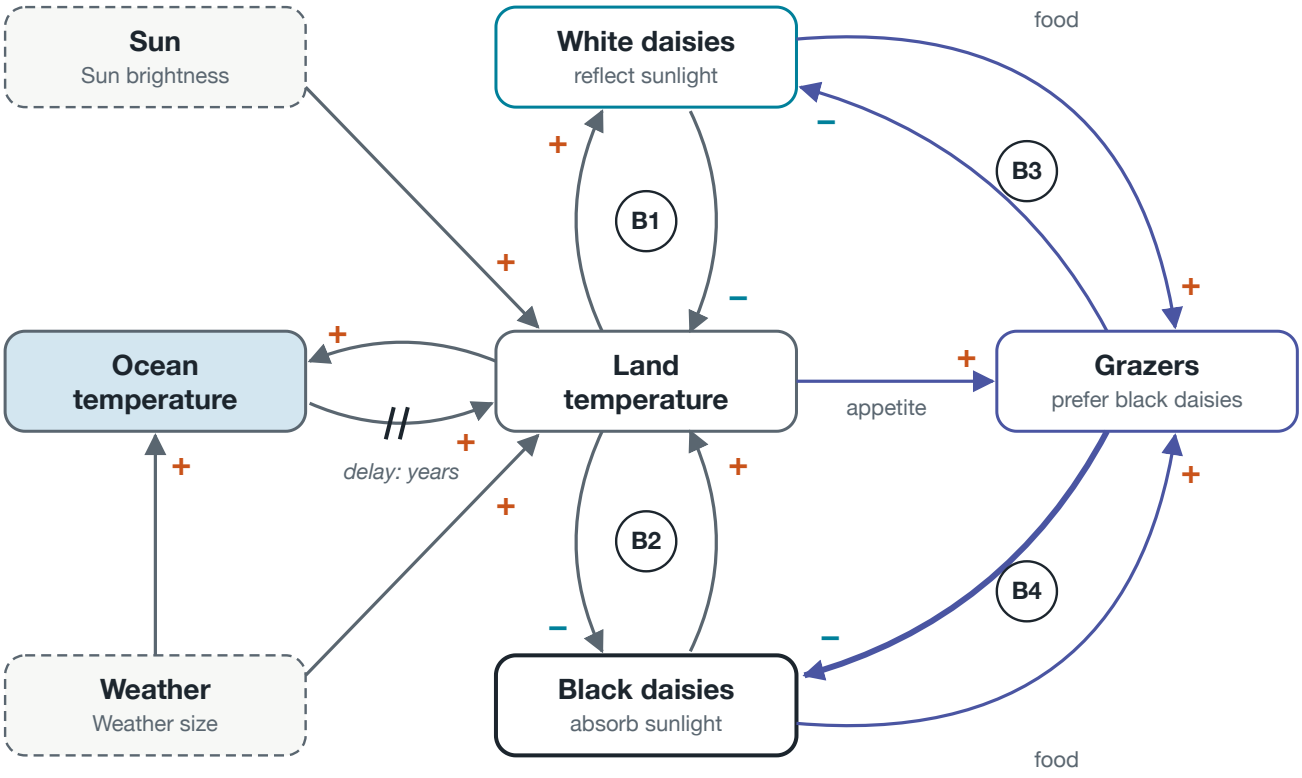
Everything from part 1 is still here: land, a mixed layer 75 m deep over 70% of the planet, a deep ocean, the sun and the weather. Life adds three things, all on the land.

- **White daisies.** They reflect more sunlight than bare ground, so a patch of white daisies is cooler than the land around it, by about 3–4 K.
- **Black daisies.** They absorb more sunlight than bare ground, so their patches are warmer than the land around them.
- Both colours grow best when **their own patch** is at 22.5 °C, more slowly either side, and not at all below 5 °C or above 40 °C. They can only spread onto bare ground, and they die at a steady rate. So white daisies do best when the land is warm, and black daisies when it is cool.
- **Grazers.** They eat daisies of both colours, but prefer black ones. They multiply when food is plentiful and starve when it runs out. They eat fastest when the land is at 25 °C and more slowly when it is colder or warmer. The land usually sits near 19–22 °C, below their best, so in this model **warmer land means hungrier grazers**.

Life touches the climate in only one way: who covers the land sets how much sunlight the land reflects. The climate touches life in only one way: temperature sets how fast each daisy grows and how fast the grazers eat.

Two differences from part 1. The life presets run with **Day and seasons** unticked: the sun is steady, and the weather is a fresh random push every quarter-day that doesn't last. They were tuned that way, so leave the box unticked for presets 4 to 7. To compare a life run with a bare planet fairly, load preset 3 and untick **Day and seasons**.

1.2 The feedbacks as a causal loop diagram



The causal loop diagram for presets 4 to 7, drawn for this handout. The "How it works" tab shows the same loop as a cycle of six stages; this diagram shows the links that make it.

How to read it.

- An arrow from A to B means "a change in A changes B".
- **+** at the arrowhead means they move the **same way**: more A, more B; less A, less B.
- **−** means they move **opposite ways**: more A, less B.
- Follow arrows round until you come back to where you started, and you have a **loop**. Count its − signs. An **odd** number makes a **balancing loop (B)**: it pushes back against a change. An **even** number, or none, makes a **reinforcing loop (R)**: it makes a change grow.
- ‖ marks a **delay**: B responds to A only after a while.
- **Dashed boxes** are outside drivers. They push on the planet, but nothing on the planet pushes back on them.

The signs hold near the land's usual temperatures, about 18–22 °C. Far from there, some can flip: on very hot land, for example, even white daisies grow worse as it warms.

The links, one loop at a time.

- **B1, the white-daisy thermostat.** Warmer land → white patches move toward their best temperature, so white daisies spread (+) → more sunlight is reflected, and the land cools (−). One − sign: balancing.
- **B2, the black-daisy thermostat.** Warmer land → black patches, already warmer than the land, get too hot, so black daisies shrink (−) → less sunlight is absorbed, and the land cools (+ from black daisies to temperature: fewer black daisies, cooler land). One − sign: balancing. Turn it round and it says what part 2 will show: **cold land favours black daisies.**
- **B3 and B4, food.** More daisies → more grazers (+, food). More grazers → fewer daisies (−, eating). One − each: balancing. But it is slow: grazers take months to multiply, and months to starve. The thick line says that grazers prefer black daisies.

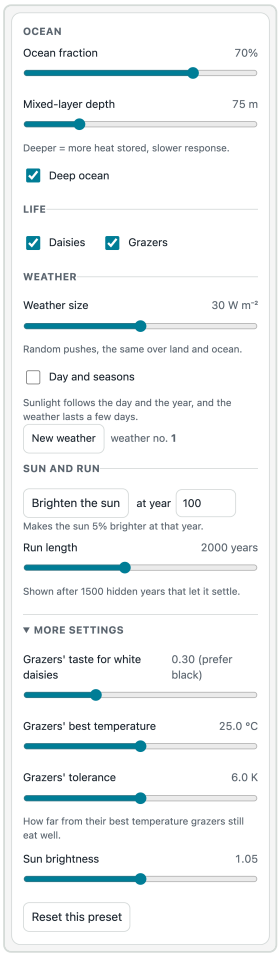
- **Appetite.** Warmer land → hungrier grazers (+). This is how the climate reaches the grazers.
- **Land and ocean.** Heat flows between the land and the ocean, both ways (+ and +). The land holds little heat and the ocean a lot, so the ocean changes slowly and the land is dragged toward the ocean's temperature, **years** late (‖). This pair is not a runaway loop: heat always flows from warm to cold. Read it as "the land is tied to a slow ocean".
- **Outside drivers.** The **sun** warms the land (and the ocean, not drawn). The **weather** pushes on the land and the ocean together, at the same moment.

Try it yourself. Find the path Land temperature → Grazers → White daisies → Land temperature. Count its signs. Is it balancing or reinforcing? When would it matter most?

It has two − signs, so it is reinforcing. Hungrier grazers strip white daisies too, the land darkens, warms, and the grazers eat faster still. During a boom, this is what makes the boom run away until the food is gone.

Why balancing loops can make cycles. A balancing loop pushes back, so you might expect it to hold the planet steady, and with no delay it does (Run 1 and Run 3). With a long delay it overshoots. Think of a shower with a long pipe: you turn it hotter, nothing happens, so you turn it more, and by the time the hot water arrives it is scalding. You turn it back, too far, and so on. The ocean is the long pipe. Grazers strip the land, the planet brightens, but the land cools only as fast as the ocean lets it, and by the time it is cold the daisies have already changed again. This is Run 2.

1.3 The controls, and what each one does



*The controls, with **More settings** open. Every change reruns the model, which takes a few seconds.*

Presets run along the top: **4 Add life**, **5 Compare: grazers, all land**, **6 Compare: grazers, no weather**, **7 Clockwork cycle**. A preset runs by itself as soon as you press it. Any slider you move turns the label into "your own settings"; **Reset this preset** (at the bottom of **More settings**) puts everything back.

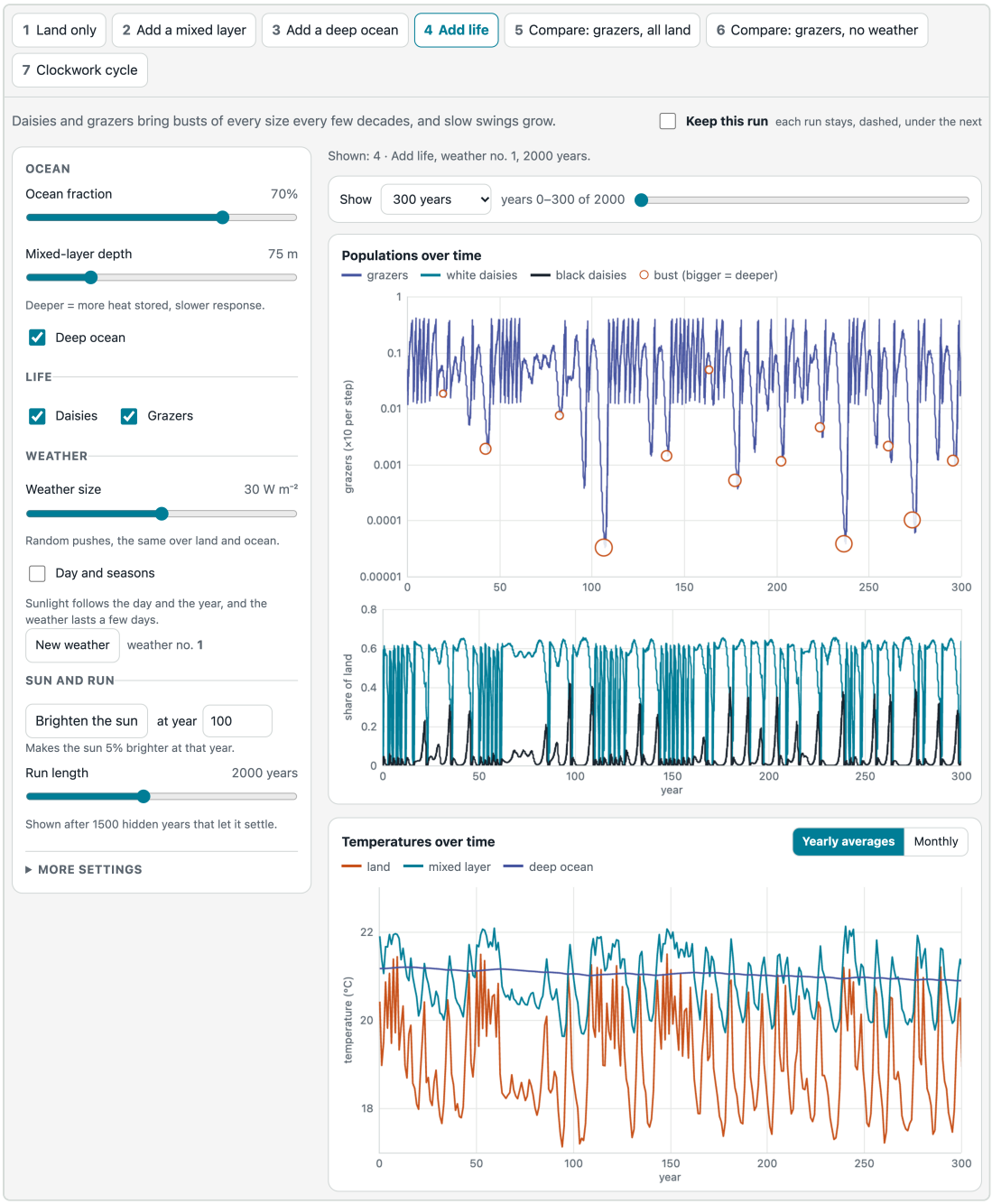
Keep this run keeps the current run on the charts, dashed, so you can compare the next run with it. Tick it before you change something.

The table says where each control sits in the diagram and what it does. The "What you'll see" column comes from runs of the app's own model; your numbers should be close.

Control	What it changes in the model	In the diagram	What you'll see
Ocean fraction	How much of the planet is ocean. 0% leaves an all-land planet with no mixed layer or deep ocean.	Removes or strengthens the ‖ delay	Preset 6 at 0%: the cycle stops and everything settles. At 30%: a faster cycle, about every 3 years, with deeper busts.
Mixed-layer depth	How deep the stirred top of the ocean is, so how much heat it holds.	Lengthens the ‖ delay	Preset 7: one turn of the cycle takes about 7 years at 25 m and about 11 years at 150 m.
Deep ocean	Adds the slow store of heat below the mixed layer.	A slower store behind the ocean	Preset 6: a slightly slower cycle with it (about 8 years) than without it (about 7.4).
Daisies	Life on or off. With it off, the land is bare ground and there are no grazers either.	Removes B1–B4	You are back to part 1's bare planet.
Grazers	Grazers on or off.	Removes B3, B4 and the appetite link	Daisies alone steady the land (Run 1).
Weather size	How big the random pushes are. 0 switches the weather off.	The size of the Weather driver	Preset 4: the century bar grows with it, from about 0.05–0.09 K at 10 W/m² to 0.25–0.38 K at 60 W/m².
Day and seasons	Sunlight that rises, sets and swings through the year, and weather that lasts a few days.	Changes the Sun and Weather drivers	Leave it unticked for presets 4–7 (see 1.1).
New weather	Reruns with a different random sequence; the weather number goes up by one.	Same drivers, different luck	The busts move to different years; the pattern stays. Use it to check that a result isn't luck.
Brighten the sun, at year …	Makes the sun 5% brighter from the year you type.	A step in the Sun driver	A sudden push. Watch which daisy colour gains.
Run length	How many years are shown, after 1,500 hidden years that let the planet settle.	–	Shorter runs are quicker. Keep 2,000 years for the century bar.
Grazers' taste for white daisies *(More settings)*	0 = they eat only black, 1 = only white, 0.5 = no preference. The presets use 0.3.	The thickness of the B3 and B4 lines	See Run 6.
Grazers' best temperature *(More settings)*	The land temperature at which they eat fastest. The presets use 25 °C (26 °C in preset 7).	The strength of the appetite link	See Run 6.
Grazers' tolerance *(More settings)*	How far from their best temperature they still eat well. The presets use 6 K (5 K in preset 7). Smaller means fussier.	The strength of the appetite link	See Run 6.

Control	What it changes in the model	In the diagram	What you'll see
Sun brightness *(More settings)*	How bright the sun is all the time, relative to today's. The presets use 1.05.	The size of the Sun driver	With daisies, the land's temperature hardly changes over a wide range (Run 1).

1.4 A tour of the screen



- **Show** picks how many years the charts show: 10, 50, 300, 1000 or the whole run. The slider next to it moves the window through the run.
- **Populations over time** has two charts. The top one shows the grazers on a **log scale**: each gridline is 10 times the one below, so a fall of two gridlines is a fall of 100 times. The bottom one shows how much of the land is covered by white and by black daisies.
- **Circles** on the grazer chart mark busts; a bigger circle is a deeper bust. Circles are kept at least 12 years apart, so when the grazers cycle faster than that, only every other dip gets one. **To time a cycle, measure from dip to dip, not from circle to circle.** Tap a circle to pick that bust: the line under the charts then says how many times the grazers fell.

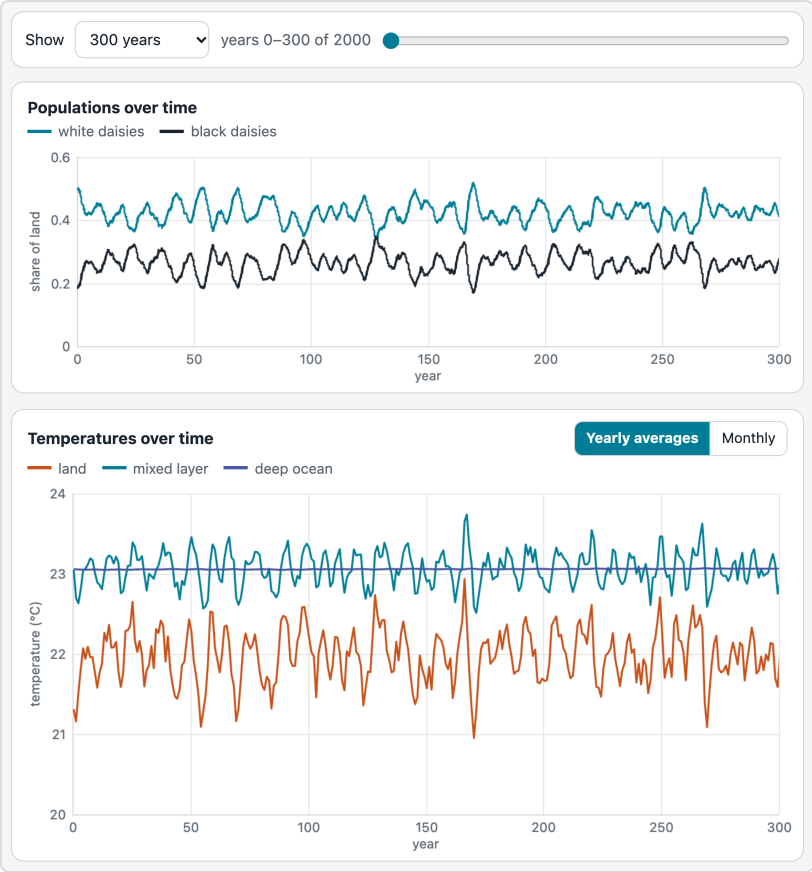
- **Temperatures over time** shows the land, the mixed layer and the deep ocean, as yearly averages or monthly values.
- **The planet** box appears when you move over or tap any chart. It shows how the planet is covered at that moment (ocean, bare ground, white and black daisies) and reads off each temperature, the grazers, how much sunlight the whole planet reflects (**planet albedo**) and the sun.
- **How much does temperature vary at each timescale?** works as in part 1: the bars show the spread of the planet's temperature after averaging over a month, a year, 10 years and 100 years. The dashed line is what weather alone would give, if each month were independent of the last.
- **Download** (below the charts) saves the run as a spreadsheet file. See "Take it further".

Part 2. Six runs to explore

Run 1. Daisies without grazers

Set: preset 4, **Add life**, then untick **Grazers**.

Before you look: in the diagram, only B1 and B2 are left. Both are balancing. Will the land's temperature wander more or less than on the bare planet? Which bar do you expect to change most?



What to look for. White daisies cover about 42% of the land, black about 26%, and both jiggle with the weather. The land sits near 22 °C, close to the daisies' favourite. Compare the bars with the bare planet (preset 3 with **Day and seasons** unticked):

Spread of the planet's temperature	Bare planet	Daisies, no grazers
over 1 year	0.23–0.25 K	0.24–0.27 K
over 10 years	0.15–0.18 K	0.07–0.09 K
over 100 years	0.05–0.08 K	0.013–0.023 K

(Ranges cover weather numbers 1 to 20. Your numbers should fall in or near them.)

The year bar hardly changes. The slow bars **shrink**, the century bar by about three times.

Now push the sun. Set **Weather size** to 0, open **More settings**, and move **Sun brightness**. Read the land's temperature in the planet box. Then untick **Daisies** and do it again.

Sun brightness	0.95	1.00	1.05	1.10	1.15
Bare land	12.4 °C	18.2 °C	23.9 °C	29.7 °C	35.5 °C
Land with daisies	19.0 °C	24.9 °C	22.0 °C	21.6 °C	27.1 °C
Daisies (white / black cover)	0 / 69%	1 / 68%	43 / 26%	70 / 0%	69 / 0%

From a sun of 0.95 to 1.10, bare land warms by 17 K; land with daisies stays between 19 and 25 °C.

Why. The two thermostats. When the land warms, white daisies gain and black lose, both of which cool it; when it cools, the reverse. From year to year the daisies make little difference, but a drift that lasts decades gives them time to answer, so they take the slow swings out. Under a dim sun the land is covered in black daisies, and under a bright one in white. Past 1.10, white daisies already cover as much land as they can, so the thermostat has nothing left to give, and the land warms again.

Run 2. Add grazers, but no weather

Set: preset 6, **Compare: grazers, no weather**. Set **Show** to 50 years.

Before you look: nothing outside the planet changes: the sun is steady and there is no weather. All the loops in the diagram are balancing. What will the grazers do over 50 years?



What to look for. A cycle that repeats for ever, one big boom about every **8 years** (the turns alternate a little, about 7 and 9 years). The grazers swing 24 times, from about 0.016 to 0.39, and the land between 18 °C and 21 °C. Every bust is the same size.

Follow one turn with the planet box, counting years after a grazer peak:

1. **Boom (0 years).** Grazers peak and strip the land, black daisies first. Bare ground is darker than white daisies, so the planet is at its warmest about 0.7 years later.
2. **Bust (0.7 years).** The food is gone; the grazers starve.

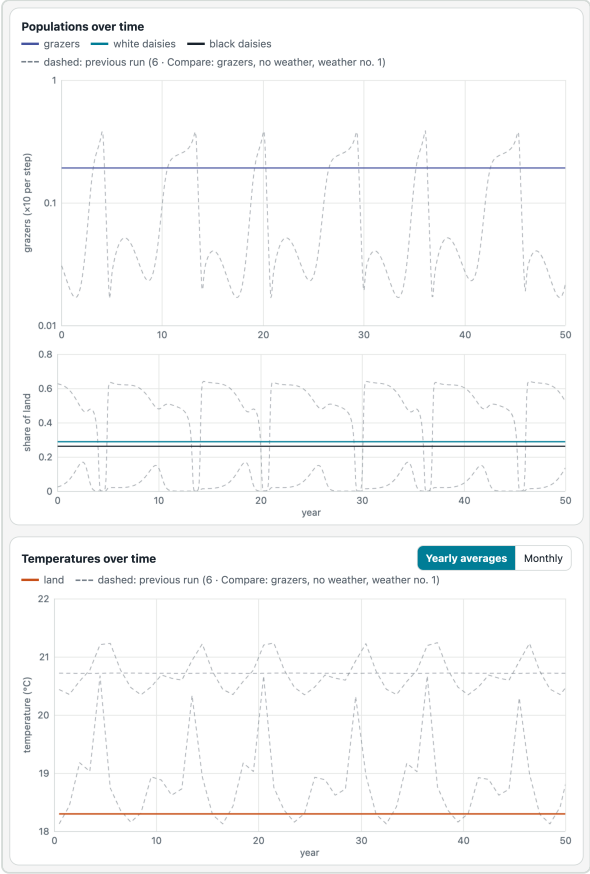
3. **White bloom (1.1 years).** Free of grazers, white daisies cover up to 64% of the land. The planet is at its brightest.
4. **Slow cooling.** The land is coldest at 3.6 years, and the mixed layer at 4.5 years: 2.5 to 3.5 years after the brightest moment. **This delay is the ocean's memory.**
5. **Cold suits black daisies, not grazers.** Black daisies peak at 5.6 years, at up to 15% cover. Grazers eat less in the cold.
6. **Rewarming and the next boom (about 8 years).** Black daisies darken the land, the planet warms, and the grazers have warmth and their favourite food: boom.

Why. No step needs anything from outside. Each balancing loop answers the last change, but through the ocean, years late, so each answer overshoots. Put your finger on the diagram and follow the turn: grazers → black daisies (B4), food runs out, white daisies take the land (B3), the land cools through the ocean's delay (‖), black daisies spread (B2), the land warms, and the grazers' appetite returns.

Run 3. Take the ocean away

Set: stay on preset 6, tick **Keep this run**, then move **Ocean fraction** to 0%.

Before you look: the same daisies and grazers, the same steady sun, no weather, and now no ocean. Will the cycle be faster, slower, or something else?



What to look for. No cycle at all. After the hidden years, everything sits still: grazers at 0.19, white daisies at 29%, black at 26%, land at 18.3 °C.

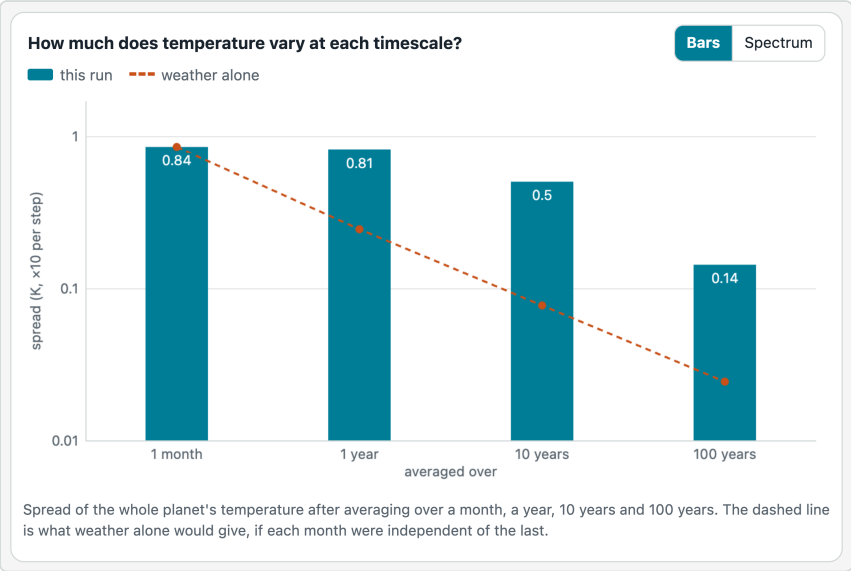
Now load preset 5, **Compare: grazers, all land**: the same all-land planet, with weather. The grazers boom and bust about **every 2 years**, and the busts are small: in a typical one the grazers fall less than 3 times.

Why. Without the ocean, there is no long delay. The land answers within weeks, so the loops push back in time and the planet settles. A kick from the weather sets it ringing for a while, like a bell, but it dies down, so the busts stay small and fast. **The ocean's delay is what keeps the cycle going.** For a check, try 30% ocean on preset 6: a cycle comes back, about every 3 years.

Run 4. Add weather: busts of every size

Set: preset 4, **Add life**. Show 300 years.

Before you look: the weather is random and forgets within a quarter-day. In Run 2 every bust was the same. Will weather make the busts all a bit bigger, all a bit smaller, or different from each other? What will it do to the century bar?



What to look for. Look at the grazer chart in the screenshot of section 1.4: the circles have every size. In a typical bust the grazers fall 8–17 times; in the smallest tenth only about 3 times; in the largest tenth 200 to 330 times. Big busts come every few decades.

Now the bars. The century bar is 0.11–0.28 K: **two to four times** the bare planet's with the same weather number (0.05–0.08 K), and roughly ten times that of the daisies without grazers in Run 1. The decade bar sits far above the dashed line.

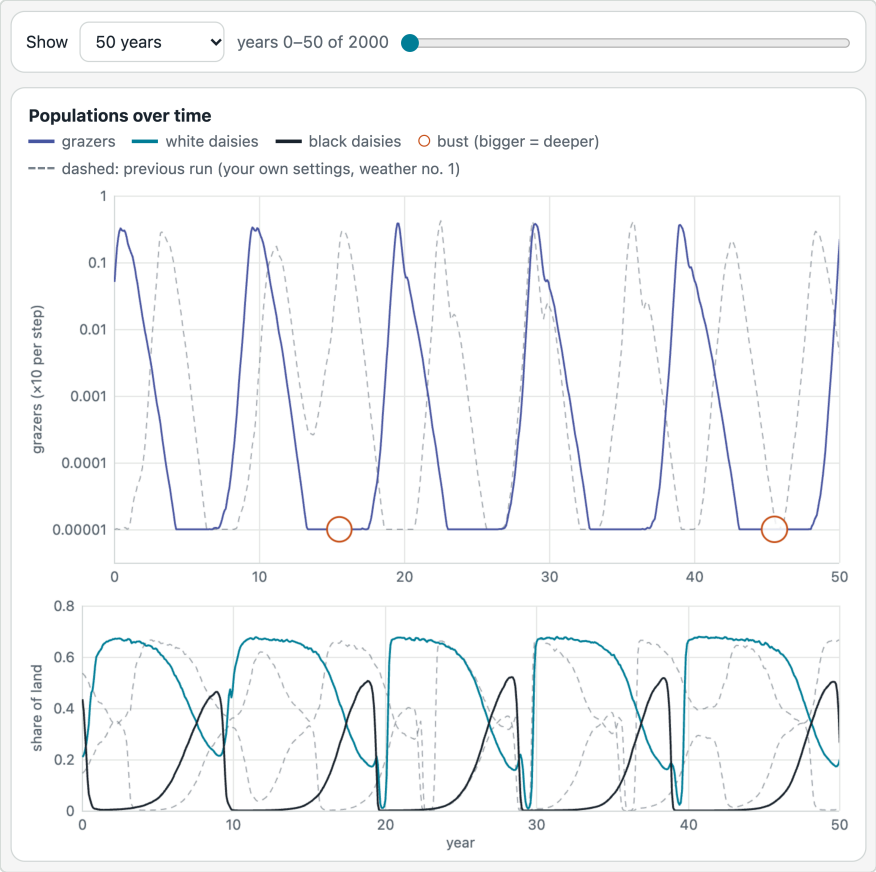
Find three busts of different sizes. Tap each circle and read the line under the charts to see how far the grazers fell. Then move along the temperature chart to the three years just before each bust, and read the **mixed layer** in the planet box. What do you notice?

Why. The weather's pushes are fast and small, but the ocean adds them up, as in part 1. So some years the ocean goes into a bust colder than usual. A colder ocean keeps the land cold for longer after the white bloom: the grazers starve for longer, and the bust goes deeper. Across many busts, the colder the ocean in the three years before, the deeper the bust (a correlation of about −0.8; −1 would be a perfect match). The loop sets **when** busts come; the ocean's memory of the weather sets **how big** each one is. Daisies alone took slow swings out (Run 1); daisies, grazers and an ocean together put far more back in.

Run 5. The ocean sets the beat

Set: preset 7, **Clockwork cycle**. Move **Mixed-layer depth** to 25 m. Tick **Keep this run**, then move it to 150 m. Show 50 years.

Before you look: in the diagram, depth sets the length of the delay ‖. A mixed layer six times deeper: will a turn of the cycle take longer, take less time, or stay the same?



What to look for. Time the cycle from one dip in the grazer chart to the next (not from circle to circle).

Mixed-layer depth	25 m	50 m	75 m	150 m
One turn of the cycle, no weather	6.7 years	7.7 years	8.4 years	10.7 years

Why. A deeper mixed layer holds more heat, so the land's cooling after each white bloom drags on for longer, and the whole loop waits for it. **A longer pipe, a slower shower.** In part 1, depth set the timescale of the ocean's wandering; here it sets the beat of the life cycle as well.

Run 6. Change the grazers

Set: preset 6 (no weather, so you can see the pattern clearly). Open **More settings** and change one grazer setting at a time. Press **Reset this preset** between changes.

Before you look: the slow, 8-year cycle needs the climate to hold the grazers back: cold land means hungry-but-slow grazers. What happens if the grazers are happy at the land's usual temperature? What if they can hardly bear it?

Change on preset 6	One turn	Grazers swing	Land
none (best 25 °C, tolerance 6 K, taste 0.3)	about 8 years	24 times	18–21 °C
best temperature 22 °C	about 1.1 years	37 times	21–23 °C
best temperature 28 °C	about 8.4 years	over 10,000 times	19–23 °C
best temperature 30 °C	grazers die out	–	steady at 22 °C
tolerance 4 K	about 7.8 years	over 10,000 times	19–23 °C
tolerance 8 K	about 1.3 years	38 times	20–23 °C
taste for white 0.5	about 1.2 years	48 times	23–24 °C
taste for white 1 (only white)	about 4 years	over 10,000 times	26–30 °C

What to look for. Two kinds of cycle. When the grazers eat well at the land's usual temperature (best temperature 22 °C, or a wide tolerance), they cycle quickly with their food, about once a year, and the land's temperature hardly moves: the grazers and daisies are cycling **without** the climate. When the cold holds them back hard (best temperature 28 °C, or a narrow tolerance), the slow, ocean-paced cycle returns, and each bust nearly wipes them out. At 30 °C they never recover, and the planet is Run 1's daisies-only planet. The taste control changes which colour the grazers clear away: with no preference the land stays mostly bare, and if they eat only white daisies, black ones take over and the land is hot.

Why. In the diagram, these controls set the strength of the appetite link and the thickness of the B3 and B4 lines. The slow cycle needs the climate in the loop. Take the climate out, and the food loops alone give the fast cycle you saw on all land in Run 3. Make the climate's grip too strong, and the grazers can't come back.

Part 3. What to take away

1. **Daisies alone are a thermostat.** Two balancing loops, through the colour of the land, hold its temperature near the daisies' favourite over a wide range of sunlight, and they take out the slow swings.
2. **A delay turns a balancing loop into a cycle.** The ocean's slow response makes the grazer–daisy–climate loop overshoot, so it cycles for ever with nothing outside cycling. Take the ocean away and it settles.
3. **The ocean sets the beat.** A deeper mixed layer gives a longer cycle.
4. **The ocean's memory sets the size of each bust.** Weather is fast and small; the ocean adds it up, and a cold ocean going into a bust makes it deep. So busts come in every size, and the planet swings over decades and centuries.
5. **The cycle needs the climate in the loop.** Grazers that don't care about temperature cycle fast with their food alone; grazers held back too hard die out.

What the model leaves out. Places: there is one land and one ocean, so no daisy can move somewhere cooler. Evolution: daisies and grazers never adapt. Earth's living things do change its climate (forests darken snowy land, plants move water into the air), but these links are much weaker than Daisyworld's, which is built to make the link strong. And no day, seasons or lasting weather in the life presets.

Questions to take to the discussion.

- Find a balancing loop in the real climate system. Does it have a long delay anywhere? Could it overshoot?
- In Run 1, daisies made the slow swings smaller. In Run 4, daisies and grazers made them larger. Which parts of the diagram make the difference?
- Before the biggest bust you found, was the ocean warmer or colder than usual? Could you have forecast that bust a few years ahead? What would you need to know?

Take it further: anatomy of a bust

(About 45 minutes, after the tutorial, in Excel, Google Sheets or Python.)

1. Load **Add life** (preset 4). Press **Download all busts (stacked)**: each bust's 40 years sit one under another, with `bust_id`, `bust_drop_factor` (how many times the grazers fell) and `whole_years_from_bust` columns. Then press **Download whole run** as well.
2. **Work with anomalies (how far above or below normal).** In the whole-run file, find the average of each of these columns over the whole run: `mixed_layer_T_degC`, `land_T_degC`, `white_cover` and `black_cover`. In the busts file, add a column for each that subtracts this average from every value. Use these anomaly columns from here on.
    - Python: `run = pd.read_csv(<the _run.csv file>)`, `df = pd.read_csv(<the _busts.csv file>)`, then for each of the four columns `df[c] = df[c] - run[c].mean()`.
3. **The average bust.** Make a pivot table with `whole_years_from_bust` as rows, and the average of the four anomaly columns, `grazers` and `grazer_appetite_per_day` as values. Chart it.
    - Python: `df.groupby("whole_years_from_bust").mean(numeric_only=True)`.
4. **Big against small.** Filter `bust_drop_factor` above 30 (big busts), then below 5 (small busts), and repeat the pivot each time.
5. **What predicts the size?** For each `bust_id`, average the mixed-layer anomaly over `whole_years_from_bust` −3 to −1. Scatter-plot it against `bust_drop_factor`, with a log scale on that axis.

Questions.

- Q1. In the average bust, put these in time order: the ocean starts to cool, white cover peaks, the grazers bottom out, black cover peaks, the ocean warms past normal.
- Q2. Do big busts start differently from small ones? Look at the ocean's temperature 5 years before.
- Q3. Does the ocean's temperature in the 3 years before a bust predict how deep the bust will be? Describe the scatter in one sentence.
- Q4. Using your answers, mark each arrow you used on the causal loop diagram: which happened first?